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Research paper

Chronic effects of traumatic brain injury and the impact of biperiden treatment in a male rat model

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ABSTRACT

Traumatic brain injury (TBI) disrupts brain function and may lead to post-traumatic epilepsy (PTE), a long-term complication. Anticholinergic drugs, such as biperiden, have shown potential in modulating neural plasticity. This study evaluated whether biperiden treatment influences motor and cognitive recovery, seizure susceptibility, and neurovascular and inflammatory responses in rats subjected to the lateral fluid percussion injury (LFPI) model. Adult male Wistar rats underwent LFPI or sham surgery and were treated with saline or biperiden (8 mg/kg) administered intraperitoneally every 8 h for 10 days, starting 6 h after trauma. Neuromotor battery tests were assessed over 28 days, spatial memory at 30 days using the Barnes Maze, and seizure susceptibility was tested with pentylenetetrazol (PTZ) at 90 days after trauma as a second hit. Plasma samples were collected 10 days after trauma for Single Molecular Array (SIMOA) analysis of trauma biomarkers, neurofilament light chain (NfL), and total tau. Histological and immunofluorescence analyses were conducted at 91 days. TBI significantly increased plasma NfL levels compared to the Naive group, indicating neuronal damage. Animals that underwent LFPI and were treated with saline solution were called TBI-SAL group, while animals that underwent LFPI and were treated with biperiden solution were called TBI-BIP group. TBI-SAL group exhibited memory impairments and random search strategies in the Barnes Maze. Both TBI groups (those treated with saline and with biperiden) exhibited higher seizure susceptibility to PTZ at 30 mg/kg, but biperiden slightly reduced seizure intensity. Histological analyses revealed larger lesion volumes, reduced NeuN expression in hippocampal regions (CA1, CA3), and altered astrocytic and microglial morphology in the TBI-SAL group. Notably, biperiden treatment promoted vascular remodeling, characterized by an increase in vessel density, in the motor cortex and hippocampal regions, suggesting potential modulation of post-traumatic neuroinflammation. Biperiden enhanced vascular remodeling and partially mitigated long-term histopathological changes after TBI, though its protective effects on cognitive and seizure-susceptibility responses were limited. These findings highlight biperiden's potential to influence vascular and inflammatory responses in the injured brain.

1. Introduction

Traumatic brain injury (TBI) is a physical insult to the central nervous system (CNS) that disrupts its normal function (Ferguson et al., 2017). Neurological damage in TBI is not limited to the immediate effects of the primary injury; it may also progressively evolve through secondary mechanisms of injury. Secondary brain injury is primarily driven by brain swelling, which increases intracranial pressure and reduces cerebral perfusion, leading to ischemia and is linked to inflammation and behavioral alterations (Ghajar, 2000).

A life-long complication of TBI is post-traumatic epilepsy (PTE), characterized by one or more unprovoked seizures that occur at least a week after TBI (Jennett, 1974). In PTE, epileptogenesis is a continuing process in which a normal brain, after a head injury incident, exhibits hypersynchronous excitability (Golub and Reddy, 2022). The risk of developing PTE in individuals with TBI is correlated with the severity of

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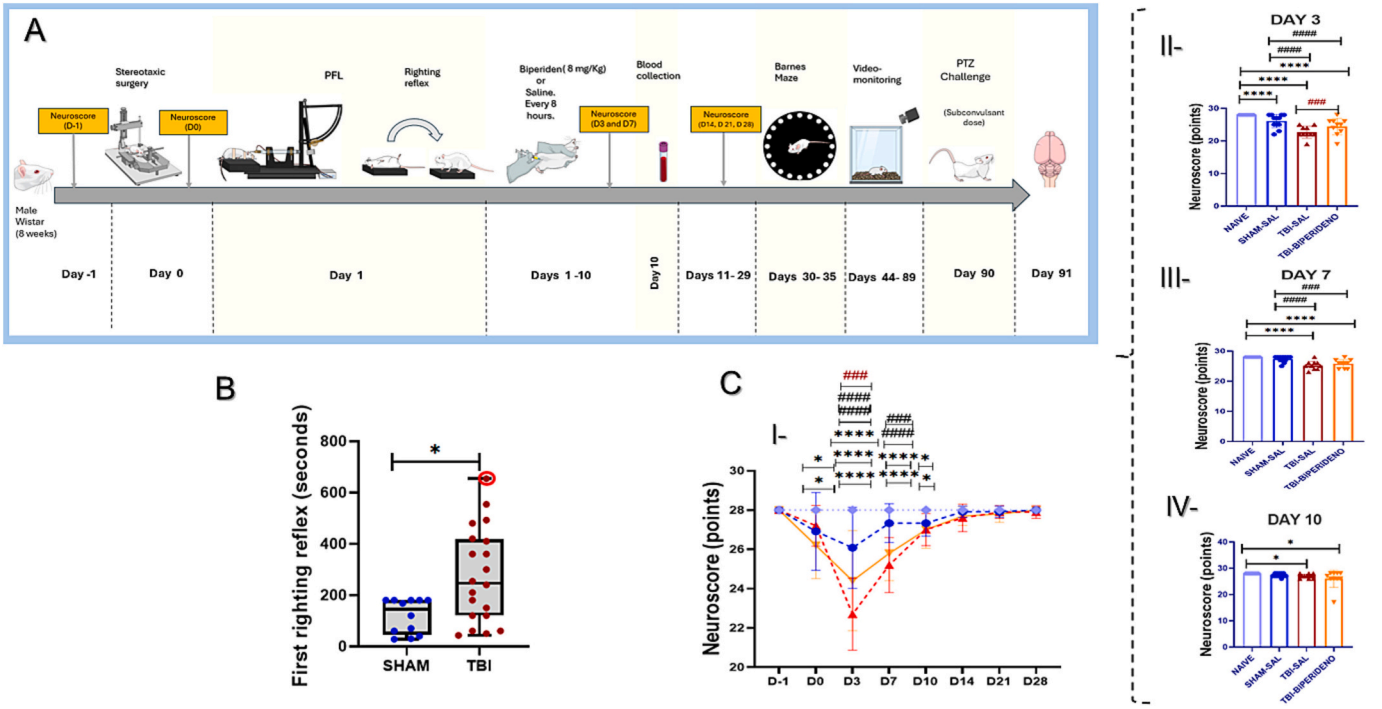


Fig. 1. Experimental Design and Validation of Moderate Traumatic Brain Injury. A. Experimental design overview. B. Comparison of the righting reflex between animals of SHAM and TBI groups. A single animal experienced trauma more severe than the intended moderate injury yet survived; this subject is marked with a light red circle. C. Composite neuroscore test (in points) conducted pre-surgery (Day −1), before the impact (Day 0), and during the recovery phase (Days 3,7, 10, 14,21, and 28). A magnified view of Day 3 is shown in panel C-II, of Day 7 in panel C-III, and of Day 10 in panel C-IV.Statistical significance: *, $p < 0.05$; **, $p < 0.01$; ***, $p < 0.001$; ****, $p < 0.001$ (compared with the NAIVE group)); #, $p < 0.05$; ##, $p < 0.01$ (compared with the SHAM-SAL group). Data are shown as the mean ± standard deviation of the mean. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

the trauma; in mild cases, the risk is estimated at 2.1 %; in moderate cases, 4.2 %; and in severe cases of TBI, the estimated risk is 16.7 % (Lowenstein, 2009).

Neuroinflammatory and neurodegenerative processes persisting and progressing weeks and months after injury provide a potential window for therapeutic intervention. Unfortunately, there are no known effective treatments for PTE (Liou et al., 2020). Although antiseizure drugs prevent the occurrence of early seizures, they are ineffective in preventing the development of PTE (Huo et al., 2022).

Preclinical studies indicate that anticholinergic drugs may modulate neural plasticity and inhibit epileptogenesis in animal models of status epilepticus-induced epilepsy (Benassi et al., 2021). Anticholinergics, such as scopolamine, have shown promise when administered shortly after brain injury (Lyeth et al., 1988). Similarly, biperiden, another cholinergic antagonist, has been shown to delay seizure onset and reduce the frequency and intensity of spontaneous seizures (Bittencourt et al., 2017; Foresti et al., 2022; Braga et al., 2025), making it a compelling candidate for further study.

Among the TBI experimental models, the lateral fluid percussion injury (LFPI) is the most commonly used to study PTE, as it closely mimics the biomechanical responses in human closed-head injuries (McIntosh et al., 1989). This model is characterized by the occurrence of spontaneous and recurrent epileptic seizures, as well as augmented brain excitability (Bolkvadze and Pitkänen, 2012). Therefore, in the present study, we investigate the effects of biperiden and TBI on memory, motor function, susceptibility to acute seizures, and neuronal damage biomarkers in rats subjected to the LFPI model.

2. Material and methods

2.1. Animals

Sixty-four male Wistar rats (8 weeks old, weighing between 230 and 350 g) sourced from the CEDEME- Center for the Development of Animal Models at the Universidade Federal de São Paulo (UNIFESP) were housed at room temperature (21 ± 2 °C), 12:12 light/dark cycle (lights on at 6:00 a.m.), humidity 45–55 %, with free access to food and water. Since arrival, animals were housed in Plexiglas hanging cages (4 animals per cage). Animal cages were cleaned and changed twice weekly and assigned randomly to the experimental groups. The animals' body weight was recorded once a week. The use of male rats in this study is justified by the 5:1 male-to-female ratio of TBI incidence in young human adults, where males are disproportionately affected (Tamás et al., 2019).

All animal handling and experimental procedures were approved by the Board for Ethics in the Use of Animals (CEUA, Comissão de Ética no Uso de Animais), an institutional ethics committee of UNIFESP, protocol number 7641161222, and were conducted in accordance with the ARRIVE guidelines for animal care and use of laboratory animals. All efforts were made to minimize the number of animals and their suffering.

2.2. Model of TBI: lateral fluid percussion injury (LFPI)

Animals were anesthetized with ketamine (80 mg /kg; i.p.) and xylazine (10 mg/kg; i.p.), followed by intradermal injection of lidocaine (10 mg/kg;i.p.) for local anesthesia. A 10 mm midline incision was made to expose the skull. A 2 mm burr hole was drilled using an electric drill (Stoelting, Wood Dale, IL) at −3 mm anterior-posterior (AP) to bregma, and − 2.76 mm medial-lateral (ML), ensuring the integrity of the dura mater. At the craniotomy site, a cannula (a female Luer lock) was fixed with dental cement (see supplemental material for careful procedures).

Twenty-four hours after cannula fixation, animals received inhalation anesthesia (3 % isoflurane with maintenance in 2 %, at a flow rate of 1–1.5 L/min in freely breathing oxygen). The female Luer lock on the animal's skull was attached to the male Luer lock of the fluid percussion

device (Model FP301 Signal Conditioner, AmScien Instruments).

The LFPI model was applied at an 18° angle with a brief (10–15 ms) transient fluid pressure pulse impact against the exposed dura. Pulse pressures were measured by an extracranial transducer and recorded on a storage oscilloscope. Regarding the pressure impact measured in atm, animals subjected to TBI received a mean impact of 2.33 ± 0.27 atm (Fig. S1) to induce moderate LFPI (O'Dell et al., 2000); however, three animals received a severe mean impact of 2.88 ± 0.24 atm (range of 2.7–3.15 atm) (See Fig. 1S).

Following the LFPI impact, the recovery of the righting reflex (the time it takes for animals to return to a four-legged support position from a supine position) was assessed by placing the animal on its back immediately after the LFPI impact and measuring the time it took to right itself. Then, animals were returned to their home cages. Sham animals underwent identical procedures but did not receive the transient fluid percussion pulse. Animals were then randomized into NAÏVE (intact), SHAM-SAL (vehicle), TBI-BIP, or TBI-SAL groups. Mortality rate and righting reflex were included to validate the TBI model and to confirm that the injury severity was consistent.

2.3. Experimental design and groups

The experimental design proceeded, as shown in Fig. 1A. All animals were distributed into four groups:

- NAÏVE: Intact animals that underwent neither craniotomy nor LFPI impact and did not receive any treatment except for pentylenetetrazol (PTZ) doses ($n = 15$).
- SHAM-SAL: Animals underwent craniotomy but did not receive the LFPI impact. They were treated with saline solution injections via intraperitoneal (i.p.) and received PTZ doses ($n = 12$).
- TBI-SAL: Animals underwent LFPI, were treated with saline solution injections (i.p.), and received PTZ doses ($n = 10$).
- TBI-BIP: Animals underwent LFPI, were treated with biperiden injections (i.p.), and received PTZ doses (n = 10).

2.4. Biperiden treatment

Animals were treated with biperiden hydrochloride dissolved in 0.9 % saline at a concentration of 8 mg/Kg, administered intraperitoneally (Cristália, Brazil). Saline (0.9 %) served as a vehicle. Treatment started 6 h after the induction of TBI or 6 h after the induction of anesthesia in the sham-operated rats based on Benassi and colleagues' (2021) protocol (Benassi et al., 2021). Treatment was continued at 8-h intervals for ten days following the same protocol (Benassi et al., 2021). The anti-epileptogenic dose of biperiden had been previously standardized by Gorgati (2009) (Gorgati, 2009). Furthermore, Benassi and colleagues (Benassi et al., 2021) also demonstrated the dose-dependent effects of a similar anticholinergic treatment, confirming its safety when administered post-TBI.

2.5. Single Molecular Array (SIMOA)®

Ten days after trauma, animals (n = 6–9 per group) were anesthetized with 4 % isoflurane for induction and 2 % for maintenance (flow rate: 1–1.5 L/min in oxygen, spontaneous breathing). Following Rodrigues et al., 2017 protocol (Rodrigues et al., 2017) animals were positioned in dorsal decubitus, and the lower lip was retracted to expose the gingiva. Using a 26G needle attached to a 1.0 mL syringe washed with heparin, the mucosa immediately below the gingival–incisor junction was punctured (depth: 2 mm; angle: 30°–60°, caudally, along the midline).

Approximately 800 μL of blood was collected per animal into micro-EDTA tubes, gently inverted, and centrifuged at 3,000 rpm for 10 min at room temperature using a Baby centrifuge (FANEM model 206). Plasma was aliquoted into 200 μL portions and stored at −80 °C for SIMOA analysis. After sampling, sterile gauze was applied for 1 min with pressure maintained until bleeding ceased. Animals were kept on a heating plate until full recovery. All procedures used aseptic techniques and disposable materials.

Plasma biomarkers related to TBI and epileptogenesis—total tau (T-tau, neuronal function) and Neurofilament Light Chain (NfL, neuronal damage)—were quantified using the Neurology 4-Plex B (N4PB) assay (Quanterix, Lexington, MA) as described by Braga et al., 2025 (Braga et al., 2025). Samples were thawed, vortexed, and centrifuged, and then 60 μL was pipetted into each well of a 96-well plate with assay calibrators. The SIMOA HD-1 (Richet/IDOR, Rio de Janeiro) was loaded with beads, reagents, discs, and plates, then run for 4–5 h. Automated cycles performed calibration, pipetting, washing, incubation, and signal detection. Fluorescent intensity (pg/mL) was converted to biomarker concentrations via a calibration curve generated by the SIMOA software.

2.6. Composite neuroscore

Sensorimotor function was evaluated before (days −1 and 0) and after TBI (days 3, 7, 10, 14, 21, and 28) using a battery of subtests comprising the composite Neuroscore test, as previously described (McIntosh et al., 1989; Andrade et al., 2022). The test involves scoring each animal on a scale ranging from 0 (severely impaired) to 4 (normal strength and function) for each of the following modalities: 1) right and left forelimb flexion during the tail suspension (4 points each), 2) right and left hindlimb flexion while the rat was gently pulled back by the tail along the table surface (4 points each), 3) ability to resist right and left lateral pulsion (4 points each), and 4) ability to stand on an inclined 45-degree angle board (4 points). The maximum score was 28. Baseline composite neuromotor scores were calculated 24 h before the stereotaxic surgery (Day −1). Moreover, it was measured again after the stereotaxic surgery before the TBI injury (Day 0). The measurement taken on D0 allowed us to distinguish the effects of the surgical procedure from those of the TBI. Furthermore, the recovery of neuromotor function was evaluated by two blinded researchers at days post-injury 3, 7, 10, 14, 21, and 28. All tests were performed between 8 a.m. to 9 a.m.

2.7. Barnes maze

The maze consists of a circular, black-colored platform with a 120 cm diameter, 60 cm off the ground. There are 20 holes allocated in the perimeter of the platform, each 10 cm in diameter. A small escape box is placed under the platform at one of the holes. Four visual cues, each with a different shape and colour, were placed on each of the four sides of the Barnes maze. White curtains surrounded the apparatus to avoid distracting environmental cues from the experimental room. On day 1, animals were habituated by exploring the arena for 10 min under moderate lighting, followed by 1 min in the escape box (Pitts, 2018). The first training began one hour after habituation.

Training consisted of two acquisition trials per day, two hours apart, to evaluate short-term memory over three consecutive days, performed from 10:00 a.m. to 2:00 p.m. The experimenter randomly placed the animal in one of the four quadrants covered by a starting box for 20 s and then released it to find the escape hole/box. If it does not find the escape box within 3 min, the rat is gently guided to it and kept there for 30 s before returning to the home cage. During each trial, an intense white light was set. The platform and escape box were cleaned with 20 % ethanol and water between trials and animals.

Forty-eight hours after the final session of acquisition training, animals undergo a 60-s probe trial in which the escape tunnel is removed from the apparatus to evaluate long-term memory. In the probe trial, the start chamber is placed in the center of the apparatus rather than the center of any given quadrant.

Latency to the first entrance into the escape box, distance traveled, and velocity were quantified for both the training and probe trials. The trials are recorded and analyzed using NOLDUS EthoVision XT software

(Noldus, Wageningen, the Netherlands) and manual counting. Searching strategies were also studied in the final probe trial.

2.8. Video monitoring

Forty-four days post-TBI, animals from the TBI-SAL and TBI-BIP groups were monitored 24 h over 45 days using an IR-light video camera coupled to a recording system (VD16E480C model, Intelbrás, Brazil) to detect spontaneous seizures. Two blinded researchers analyzed video recordings at maximum speed in experimental condition. NAÏVE and SHAM-SAL groups were not included in the recordings.

2.9. PTZ challenging test

To evaluate seizure susceptibility following TBI and assess intra-group differences, 90 days after the impact, animals were submitted to a PTZ challenge. Sub-convulsive doses of PTZ (10 mg/kg, i.p.; Sigma, St. Louis, MO). were administered in a ramp-up manner every 15 min up to a cumulative dose of 40 mg/kg, i.p., or or until the animal developed severe generalized seizures that required interruption of the protocol (Fig. 3B). Following the PTZ injection, each animal was placed in a transparent plastic cage and observed for 45 min. Latency to the first behavioral seizure activity (time between PTZ injection and the onset of the first tonic-clonic activity) and seizure intensity stage, according to the revised Racine scale for PTZ-induced seizures (Lüttjohann et al., 2009; Vahid et al., 2022) were recorded.

2.10. Brain tissue preparation

Twenty-four hours after PTZ-induced seizures, the animals were deeply anesthetized with pentobarbital sodium (100 mg/kg, i.p.). A sternotomy was made to access the heart. Then, the animals were perfused through the heart. Perfusion started with 100 mL of 0.1 M PBS solution (phosphate buffer solution; 5.52 g of monobasic sodium phosphate plus 21.88 g of dibasic sodium phosphate), then 100 mL of 4 % paraformaldehyde (diluted in PBS solution).

The brains were removed and post-fixed in 4 % formaldehyde, followed by cryoprotection in 30 % sucrose (diluted in PBS). They remained in this solution until they showed signs of dehydration, a procedure usually occurring within 24 h. Immediately after this period, the brains were dried and then frozen at −80 °C. Subsequently, the brains were sectioned in a cryostat (Leica CM1850) in coronal sections (40 μm thick). Then, the sections were stored in an anti-freezing solution (300 g of sucrose, 500 mL of PBS solution, and 300 mL of ethylene glycol) at −20 °C until histological and immunohistochemical processing. Brain sections were selected according to the stereotaxic atlas (Paxinos and Watson, 2004), ranging from −2.76 to −4.08 mm in the anterior–posterior axis relative to bregma (dorsal hippocampus), ensuring consistent sampling of both cortical and hippocampal ipsilateral regions for histological assessment.

2.11. Histological Nissl staining

Brain slices were mounted on gelatin-coated glass slides and stained for 8 min with 1 % cresyl violet dissolved in distilled water and filtered. Stained slides were dehydrated for 1 min in 100 %, 96 %, and 70 % ethanol, cleared in xylene for 2 min, covered with Entellan® mounting medium, and coverslipped. Slides were imaged (6 brain sections for each animal), and the lesion area on the ipsilateral side of trauma cortex was assessed among the groups: NAIVE, SHAM-SAL, TBI-SAL, and TBI-BIP. Images were analyzed and imported into Fiji, a version of the ImageJ software, (National Institutes of Health, http://imagej.nih.gov/ij). The percentage of lesion volume was calculated as per the following formula (Boyko et al., 2022):

$$Lesion\ volume = \frac{volume\ of\ contralateral\ hemisphere - volume\ of\ ipsilateral\ hemisphere}{volume\ contralateral\ hemisphere} \times 100$$

2.12. Immunofluorescence

Immunofluorescence was performed to identify microglia, astrocytes, vessels, and neuronal nuclei. Free-floating sections (6 brain slices per animal) were washed five times with PBS solution to remove all anti-freezing solution and incubated in a blocking solution (250 μL of Triton X-100, 90 mL of 0.1 M PBS, 40 μL of goat serum) for 30 min at room temperature under constant stirring. After blocking, the brain slices were incubated overnight with the following primary antibodies: polyclonal rabbit anti-Iba-1 antibody (1:1000, Wako, 019–19741) for microglia/macrophages, monoclonal mouse anti-glial fibrillary acidic protein (GFAP) antibody (1:1000, Sigma, G3893) for astrocytes, and monoclonal mouse anti-neuronal nuclei (NeuN) antibody (1:1000; Millipore, MAB377) diluted in a blocking solution.

The following day, after washing the slices in PBS, the slices were incubated for 2 h with secondary antibodies (anti-mouse Alexa Fluor™ 488, anti-rabbit Alexa Fluor™ 568, 1:600, all from Invitrogen) diluted in blocking solution. For vessel staining, after three PBS washes, the sections were incubated with tomato lectin (*Lycopersicon esculentum*; 1:200; Vector Laboratories; FL-1171-1) for 1 h. Nuclei were stained with DAPI (4′,6-diamidino-2-phenylindole, 1:10,000; Thermo Fisher Scientific, Carlsbad, CA, USA). Finally, the sections were coverslipped with Fluoromount G™ (Thermo Fisher Scientific).

The slides were analyzed using a computer-based digitizing image system, Zeiss Axiocam 503 colour (https://www.zeiss.com/axiocam), with a 20× objective. The laser and detector were maintained at constant settings for both cases during acquisition of each staining set. For each slide, cells were qualitatively evaluated for their localization and morphology in the brain areas of interest on the ipsilateral side of the trauma cortex (M1, S1, and M2) and the dorsal hippocampus (CA1, CA3, and DG). Slides and resultant images were coded, and fluorescence signals were quantified. Stained cells were quantified by integrated density, number of branches, average branch length, and triple points using Fiji (a version of ImageJ, National Institutes of Health, http://imagej.nih.gov/ij). Additionally, neural nuclei were counted using CellProfiler version 4.2.5 (https://cellprofiler.org/) (Carpenter et al., 2006). Three were randomly chosen from the six brain slices. For each region (motor cortex, CA1, CA3, and DG), three photos were taken, yielding nine pictures per animal per region and 54 images per experimental group.

2.13. Statistical analysis

The sample size (10–15 animals per strain) was determined based on previous TBI results from our group (Braga et al., 2025; Romariz et al., 2023) and existing literature. Power calculations were performed in advance using G*Power version 3.1.9.7 (Faul et al., 2007) to determine the minimum sample size required to test the study hypothesis. For behavioral tests, a priori sample size calculations were conducted using an ANOVA with repeated-measures within-between interactions. For the

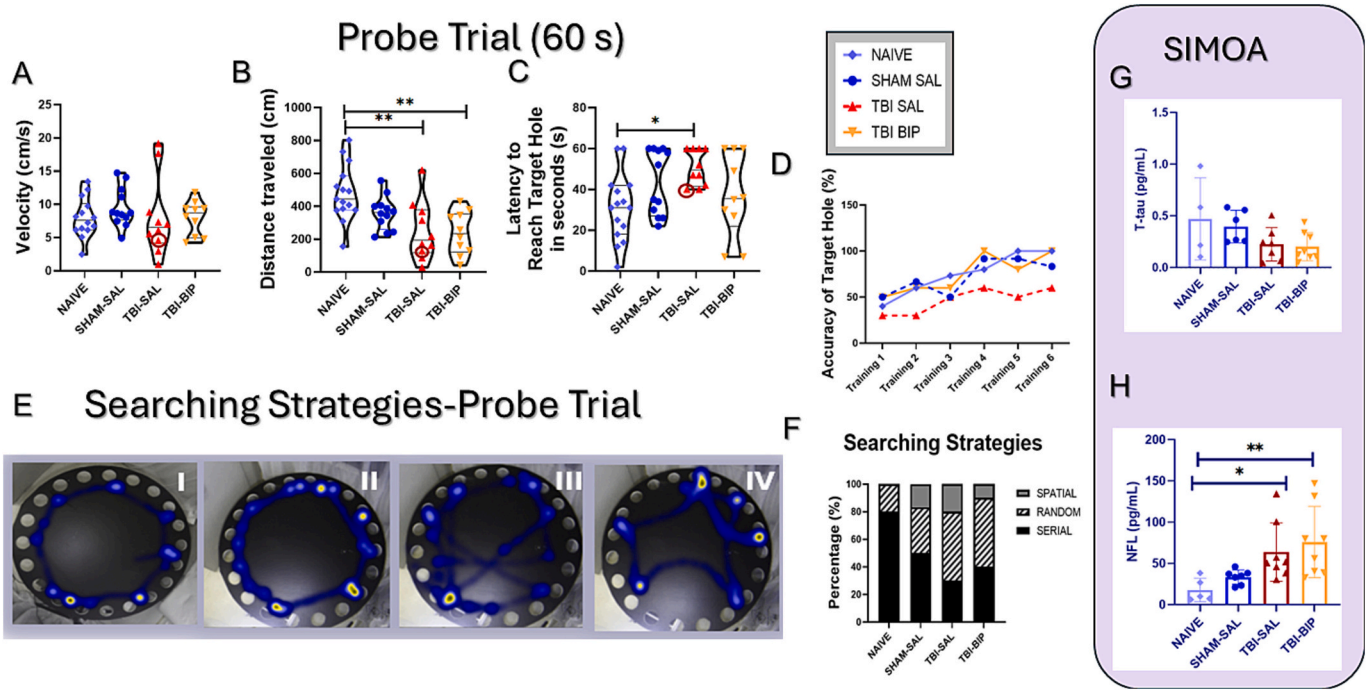


Fig. 2. Behavioral Analysis of the TBI severity and Single Molecular Array (SIMOA) analysis. A. Results of the velocity (cm/s^2) reached by all the groups during the probe trial. A light red circle represents an animal that experienced severe trauma and survived. B. Results of the distance (cm) reached by all the groups during the probe trial. Light red circles represent an animal that experienced severe trauma and survived. C. Results of the time in seconds (s) reached by all the groups during the probe trial. An animal that experienced severe trauma and survived is represented by light red circles.D. The accuracy of the animals by group in reaching the target hole. E. Heatmaps represent the search strategy of each group: I-NAÏVE, II-SHAM-SAL, III-TBI-SAL, IV-TBI-BIP. F. Percentage of searching strategy for each group.G. SIMOA analysis of total tau (T-tau) in pg/mL 10 days post trauma.H. SIMOA analysis of NfL in pg/mL 10 days post trauma. Statistical significance: *, $p < 0.05$; **, $p < 0.01$; ***, $p < 0.001$; ****, $p < 0.001$ (compared with the NAIVE group). Data are shown as mean ± standard deviation. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

composite Neuroscore battery and the Barnes maze tests, at least ten male rats were needed to establish a medium effect size of 0.50 (power = 0.90, α = 0.05).

Data analyses were performed using GraphPad Prism version 8.0 (GraphPad Software Inc., San Diego, CA, http://www.graphpad.com). Before analysis, normality was assessed using the Shapiro-Wilk test, and outliers were identified using the ROUT method (Q = 1 %) and excluded.

Depending on the data, the T-student, one-way ANOVA, or Kruskal-Wallis test was applied appropriately, followed by Tukey's or Dunn's post-hoc test. Mixed-effect models with repeated measures were used to predict a single variable using two independent variables. Data are presented as the mean ± standard deviation (SD), with statistical significance at $P < 0.05$.

3. Results

3.1. Validating the TBI model

Mortality and righting reflex assessments were used to validate the TBI model and ensure consistent injury severity across groups. Of 64 male Wistar rats, 49 underwent stereotaxic surgery (33 TBI, 16 SHAM) and 15 served as NAÏVE controls. Seven animals (3 TBI, 4 SHAM) died during anesthesia, yielding a surgical mortality of 14.3 %. Immediately post-injury, additional deaths occurred only in animals that received the trauma (TBI groups prior to any treatment) (10/30; 33.3 %), while no further mortality was observed in SHAMs (0/12). TBI rats showed a significantly prolonged righting reflex compared to SHAM-SAL controls (276 ± 184 s vs. 118 ± 67 s; U = 56.60, $p < 0.05$; Fig. 1B).

3.2. Early improvement in neuroscore test after biperiden treatment

Neuromotor performance was assessed at baseline (Day −1), pre-impact (Day 0), and during recovery (Days 3, 7, 10, 14, 21, and 28). A significant time-dependent recovery pattern was observed across groups [group: $F(4,112) = 41.90$, $p < 0.0001$; time: $F(3,232) = 43.47$, $p < 0.0001$; group×time: $F(21,232) = 10.24$, $p < 0.0001$]. On Day 3 post-impact, TBI-BIP rats showed higher neuroscores than TBI-SAL animals (24.4 ± 2.55 vs. 22.7 ± 1.86; $p < 0.001$), indicating an early, transient improvement. Significant differences were also detected between NAÏVE (28.0 ± 0.00) and SHAM-SAL (26.1 ± 2.07; $p < 0.0001$), and between SHAM-SAL and both TBI groups ($p < 0.001$). By Day 7, no difference was observed between TBI-SAL (25.2 ± 1.40) and TBI-BIP (25.8 ± 1.40; $p > 0.05$), although both remained impaired relative to NAÏVE and SHAM-SAL rats ($p < 0.001$). By Day 10, only NAÏVE animals (28.0 ± 0.00) differed from TBI groups [TBI-SAL: 27.0 ± 0.82; TBI-BIP: 27.0 ± 0.94; $p < 0.05$], with no subsequent differences through Day 28 (Fig. 1C I-III).

3.3. Biperiden modulated spatial learning but did not prevent long-term memory impairment

Training sessions were conducted on Day 30 post-impact. Detailed results are available in the supplemental material (Fig. S2 A-C). In the probe trial, conducted 48 h after the final training session, a one-way analysis revealed significant differences in the distance traveled among the groups [$F(3,43) = 7.19$; $p < 0.001$] and in latency to reach the target hole [$F(3,43) = 3.43$; $p < 0.05$]. NAÏVE animals traveled farther distances (479.5 ± 168 cm) compared to TBI-SAL (250.3 ± 178 cm; $p = 0.0026$), and TBI-BIP (238.4 ± 131 cm; $p = 0.0015$). Also, NAÏVE animals reached the target hole faster (31 ± 16.43 s; Fig. 2A-D) than TBI-SAL (50.80 ± 8.69 s; $p = 0.023$). No differences in velocity

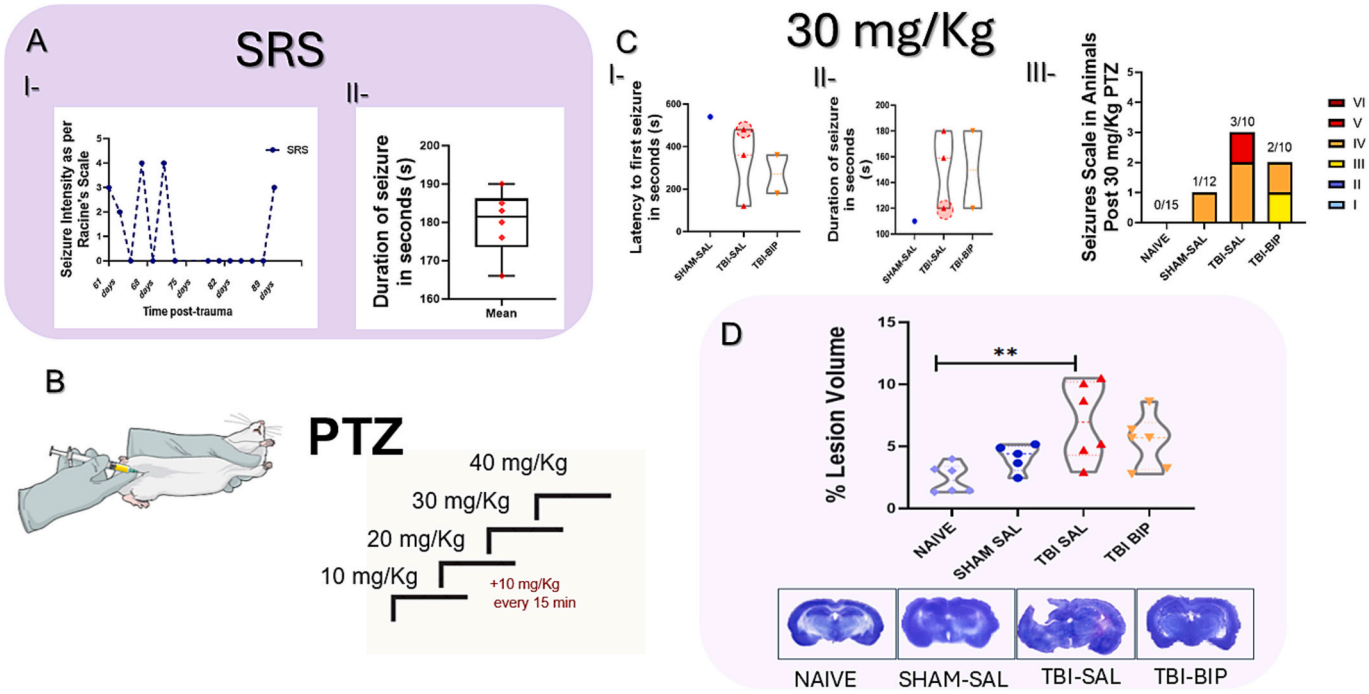


Fig. 3. Video-monitoring, PTZ challenge, and lesion volume. A. Spontaneous recurrent seizures (SRS) observed in the TBI-SAL animal with severe injury. I-. Seizure intensity reached the Racine scale level during the 45 days. II- Mean duration of seizure in seconds (s). B. Pentylenetetrazol (PTZ) scheme of the challenge in which every 15 min 10 mg/kg of PTZ was injected intraperitoneally (i.p) until the seizure. C. Results reached with 30 mg/kg of PTZ. I- Latency to first seizure in seconds (s). II- Duration of seizures in seconds (s). III- Seizure intensity according to Racine's modified scale for PTZ. An animal that experienced severe trauma and survived is represented by light red circles.D. I- Percentage of lesion volume in all groups. II- Images of Nissl representation per group. Statistical significance: *, $p < 0.05$; **, $p < 0.01$; ***, $p < 0.001$; ****, $p < 0.001$ (compared with the NAIVE group). Data are shown as the mean ± standard deviation of the mean. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

were observed among the groups.

Regarding search strategies, the NAÏVE (80 %) and SHAM-SAL (50 %) groups predominantly used a serial strategy, whereas the TBI-SAL and TBI-BIP groups relied more on random strategies (50 %). Biperiden treatment appeared to have a minor effect on strategy distribution, with a slight increase in the use of the serial strategy in the treated group (Fig. 2E-F).

3.4. Plasma NfL increased after TBI, with no significant modulation by biperiden

Plasma levels of NfL and total tau (T-tau) were measured using the ultra-sensitive SIMOA technique. For T-tau, the Kruskal–Wallis test showed no significant difference among groups (KW[4] = 5.11, $p = 0.16$), although both TBI groups (TBI-SAL and TBI-BIP) tended to have lower levels than NAÏVE and SHAM-SAL (Fig. 2G).

For NfL, the Kruskal–Wallis test indicated a significant group effect (KW[4] = 14.19, $p = 0.0019$). Dunn's post-hoc analysis revealed higher NfL levels in TBI-SAL (63.74 ± 35.38 pg/mL, $p = 0.018$) and TBI-BIP (75.86 ± 43.15 pg/mL, $p = 0.08$) compared to NAÏVE (17.80 ± 14.20 pg/mL; Fig. 2H).

3.5. Severe trauma-induced spontaneous recurrent seizures (SRS) in the videomonitoring phase

During the 45 days of video monitoring, one rat in the TBI-SAL group that received the most severe impact (3.15 atm) presented 7 seizures, with an average duration of 180 ± 8.3 s (coefficient of variation: 4.6 %; Fig. 3A). The first episode occurred 61 days post-impact, with seizure intensity progressing from stage III to V according to the revised Racine's scale (Lüttjohann et al., 2009). No SRS events were observed in any biperiden-treated rats.

3.6. PTZ challenge as a second-hit showed a trend at 30 mg/kg

Ninety days after trauma and after the 45-day video monitoring period, PTZ was administered to induce a second hit and to address whether biperiden could change the seizure threshold. No seizures occurred at 10 mg/kg, i.p., or 20 mg/kg, i.p. of PTZ. At 30 mg/kg, i.p., tonic-clonic seizures were observed in SHAM-SAL and TBI groups (TBI-SAL and TBI-BIP). At 40 mg/kg, i.p., all groups, including NAÏVE, exhibited tonic-clonic seizures, as shown in Table S1.

At 30 mg/kg, no significant differences were found between the SHAM and TBI groups regarding seizure latency or duration (Fig. 3C I-II). SHAM-SAL had a latency of 540 ± 0.00 s (seizure duration:110 ± 0.00s), while TBI-SAL had a latency of 320 ± 183 s (seizure duration:153 ± 30.4 s) and TBI-BIP had a latency of 270 ± 127 s (seizure duration: 150 ± 42.4 s).

Seizure intensity at 30 mg/kg also followed a trend in which the TBI-SAL group exhibited the highest seizure intensity (type V on the modified Racine scale for PTZ) while the SHAM-SAL reached level VI. Notably, the TBI-BIP group exhibited the lowest seizure intensity, only reaching level III (Fig. 3C). Among the 47 animals challenged with PTZ (30 mg/kg), seizures were observed in only six (12.8 %). Of these, 16.7 % occurred in the SHAM-SAL group, 50 % in the TBI-SAL group, and 33.3 % in the TBI-BIP group, indicating a higher susceptibility to seizures in TBI-SAL animals.

At 40 mg/kg, the analysis of seizure behavior also revealed no statistically significant differences among the NAÏVE, SHAM, and TBI groups in seizure latency or duration, and no clear trend was observed among the groups (see Supplemental Material Table S1 and Fig. S3).

3.7. Lesion volume was higher in the TBI-SAL group

When we analyzed the percentage of lesion volume with the Nissl

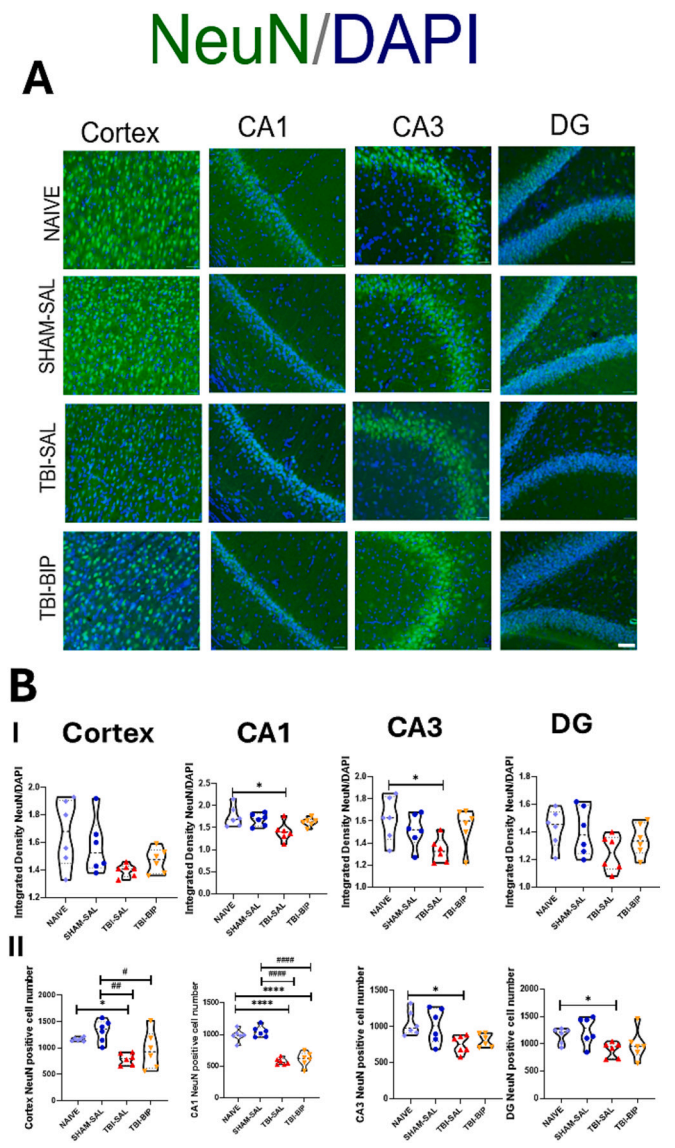


Fig. 4. Neuronal nuclei (NeuN) results. A. Photomicrographs of NeuN immunofluorescence of all groups' motor cortex and hippocampal regions (CA1, CA3, and DG). Neun (green), DAPI (blue). B I- Measure the integrated density of NeuN, correcting using DAPI, in the cortex, CA1, CA3, and DG. II- Count of NeuN-positive cells in cortex, CA1, CA3, and DG. Statistical significance: *, $p < 0.05$; **, $p < 0.01$; ***, $p < 0.001$; ****, $p < 0.001$ (compared with the NAIVE group).#, $p < 0.05$; ##, $p < 0.01$; ###, $p < 0.001$; ####, $p < 0.001$ (compared with the SHAM-SAL group). Data are shown as mean ± standard deviation. Scale bars = 40 μm. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

staining among the different groups, we observed a significant difference [F (3, 15) = 5.19; $p < 0.01$] only between the NAÏVE (% of lesion 2.40 ± 1.14) and the TBI-SAL (% of lesion 7.02 ± 3.14) groups (Fig. 3D).

TBI-SAL reduced Neuronal Nuclei expression, while biperiden treatment attenuated, but did not fully prevent it.

Neuronal nuclear protein (NeuN) immunolabeling was used to quantify neuronal loss in affected regions (Fig. 4A). One-way ANOVA revealed significant differences in NeuN expression in the CA1 [F(3,20) = 4.06, $p = 0.029$] and CA3 [F(3,20) = 3.36, $p = 0.039$]. NeuN intensity was reduced in TBI-SAL rats compared to NAÏVE in both CA1 (1.41 ± 0.21 vs. 1.73 ± 0.24, $p = 0.021$) and CA3 (1.33 ± 0.11 vs. 1.62 ± 0.20, $p = 0.037$; Fig. 4B-II).

Significant reductions in neuronal nuclei counts were observed across all regions (motor cortex, CA1, CA3, DG). In the motor cortex [F (3,20) = 6.93, $p = 0.002$], TBI-SAL rats showed a 54 % decrease versus SHAM-SAL (791 ± 109 vs. 1317 ± 209, $p = 0.002$), while TBI-BIP showed a 27 % reduction ($p = 0.04$). TBI-SAL also differed from NAÏVE ($p = 0.033$). In the CA1 [F(3,20) = 45.36, $p < 0.0001$], both TBI groups (TBI-SAL: 573 ± 52; TBI-BIP: 610 ± 113) exhibited lower neuronal counts than NAÏVE (988 ± 96) and SHAM-SAL (1045 ± 86; $p < 0.01$). In CA3 [F(3,20) = 4.48, $p \leq 0.05$], TBI-SAL showed fewer neuron nuclei than NAÏVE (754 ± 123 vs. 1044 ± 174; $p = 0.032$). Even though TBI-BIP did not show statistically significant differences compared with the other groups in CA3, it had 7.3 % more neural nuclei (809.3 ± 91.3) than TBI-SAL. Similarly, in the DG [F(3,20) = 3.40, $p = 0.037$], TBI-SAL had reduced counts versus NAÏVE (893 ± 141 vs. 1151 ± 142; $p = 0.043$; Fig. 4B-III).While TBI-BIP showed no statistically significant differences from the other groups in the DG, it exhibited a 9.6 % increase in neuronal nuclei (979 ± 268) compared with TBI-SAL.

3.8. SHAM and TBI groups have differential microglial and astrocytic expression compared to NAÏVE controls

Microglial (Iba-1) and astrocyte (GFAP) expression (integrated density) was assessed across groups and brain regions (Fig. 5A-B). One-way ANOVA showed significant differences in Iba-1 and GFAP expression in the motor cortex and DG. In the motor cortex, significant differences were observed in both Iba-1 [F (3,20) = 27.42; $p < 0.0001$] and GFAP [F (3, 20) = 10.12; $p < 0.001$] expression. Both markers exhibited increased expression in SHAM-SAL ($p < 0.0001$), TBI-SAL ($p < 0.0001$), and TBI-BIP ($p < 0.0001$) groups compared to the NAÏVE group. In the DG, significant differences were found only in the GFAP expression [F (3, 20) = 3.34; $p = 0.039$], indicating reduced GFAP expression in TBI-SAL (0.69 ± 0.16) compared to TBI-BIP (0.99 ± 0.10; $p = 0.02$)(Fig. 5B and supplemental material Figs. S4-S7).

3.9. Microglia Branching: TBI groups have fewer but longer branches

Iba-1 skeleton analysis revealed significant group differences in microglial branching in the motor cortex [F(3,20) = 7.80, $p = 0.0012$], CA1 [F(3,20) = 4.71, $p = 0.012$], and DG [F(3,20) = 5.13, $p = 0.008$] (Fig. 5C). In the motor cortex, branch numbers were reduced in TBI-SAL (111 ± 16) and TBI-BIP (100 ± 78) compared with NAÏVE (304 ± 165; $p < 0.01$).

Significant differences in average branch length were also found in the motor cortex [F(3,20) = 4.55, $p = 0.0138$] and CA1 [F(3,20) = 4.40, $p < 0.001$]. In the cortex, TBI-BIP branches were longer (8.35 ± 853 μm) than NAÏVE (5.82 ± 1.69 μm; $p = 0.009$). In CA1, branch length increased in TBI-SAL (7.821 ± 812 μm) and TBI-BIP (7.987 ± 1.05 μm) relative to NAÏVE (4.85 ± 2.01 μm) and SHAM-SAL (5.60 ± 1.06 μm; $p < 0.05$; Fig. 5C I–IV). For details, see supplemental Figs. S4–S7 and the triple-point analysis in the supplemental material.

3.10. Astrocytes branching: TBI Groups have fewer but longer branches

GFAP skeleton analysis revealed significant group differences in astrocyte branching across regions: motor cortex [F(3,20) = 7.80, $p = 0.0012$], CA1 [F(3,20) = 16.00, $p < 0.0001$], CA3 [F(3,20) = 8.37, $p < 0.001$], and DG [F(3,20) = 2.59, $p < 0.0001$] (Fig. 5D). In the motor cortex, branch number was reduced in TBI-SAL (111 ± 16) and TBI-BIP (100 ± 20) versus NAÏVE (304 ± 165; $p < 0.01$). In CA1, SHAM-SAL (306 ± 101; $p = 0.01$) exceeded NAÏVE (165 ± 86), whereas TBI-SAL (72 ± 41) and TBI-BIP (60 ± 15) showed marked reductions ($p < 0.0001$). CA3 and DG also exhibited fewer branches in SHAM-SAL, TBI-SAL, and TBI-BIP compared with NAÏVE ($p < 0.01$; Fig. 5C I–IV).

For average branch length, significant differences were detected in the motor cortex [F(3,20) = 60.55, $p < 0.0001$], CA1 [F(3,20) = 2.16, $p < 0.0001$], and DG [F(3,20) = 4.69, $p < 0.05$]. In the motor cortex and CA1, branch length increased in both TBI-SAL (14.23 ± 771 μm; 17.58 ± 4.68 μm) and TBI-BIP (12.73 ± 2.10 μm; 17.39 ± 2.43 μm) relative to

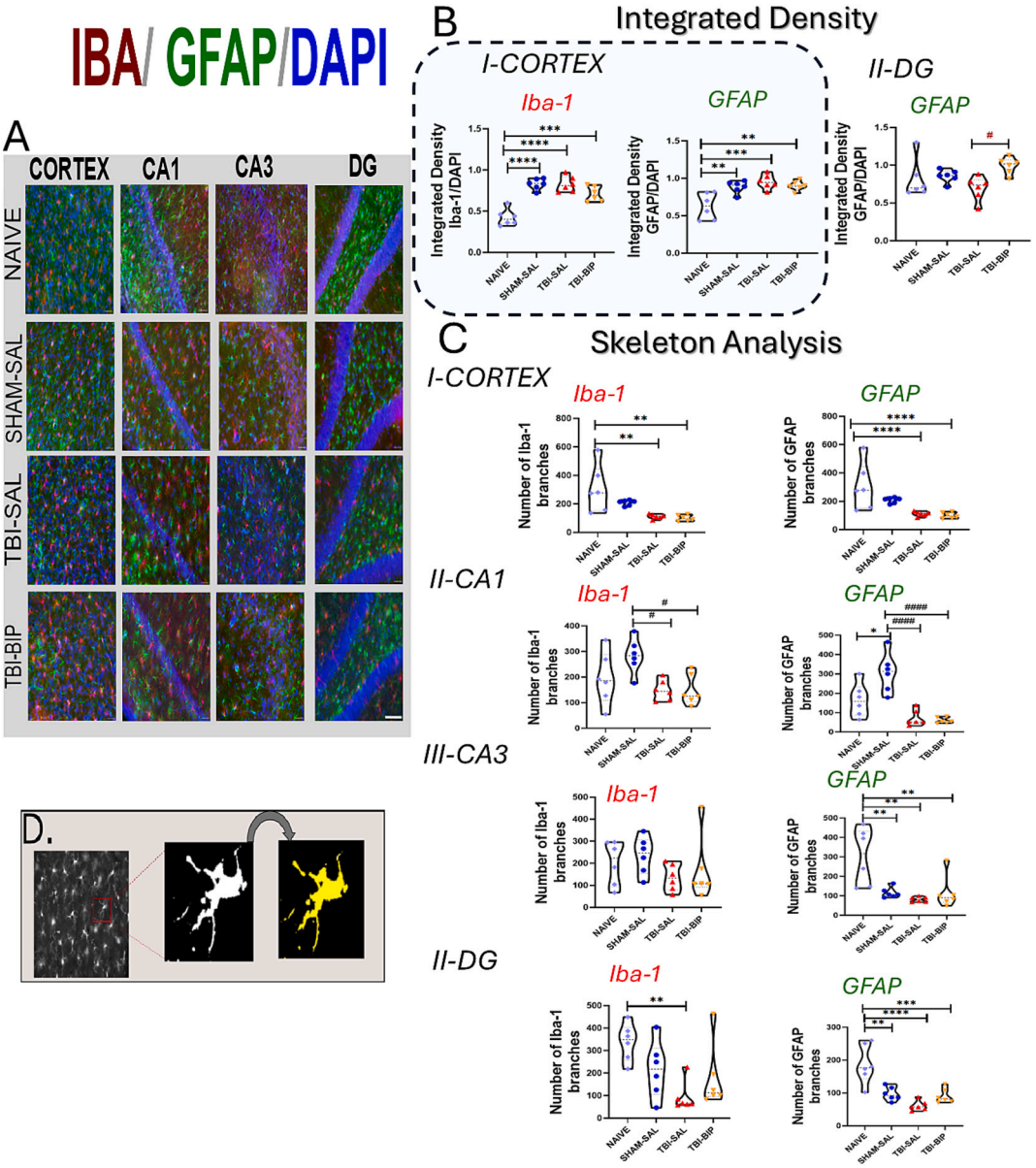


Fig. 5. Microglia (Iba-1) and Astrocytes (GFAP) results. A. Photomicrographs of Iba-1 and GFAP immunofluorescence of all groups' motor cortex and hippocampal regions (CA1, CA3, and DG). Iba-1 (red), GFAP (green), and DAPI (blue). B. Measure the integrated density of Iba-1 and GFAP, correcting using DAPI, in the cortex and DG. C. Skeleton analysis for Iba-1 and GFAP in the I-cortex, II-CA1, III-CA3, and IV- DG). Measure the number of branches (μm). D. Image representation of the skeleton analysis for astrocytes. Statistical significance: *, $p < 0.05$; **, $p < 0.01$; ***, $p < 0.001$; ****, $p < 0.001$ (compared with the NAIVE group).#, $p < 0.05$; ##, $p < 0.01$; ###, $p < 0.001$; ####, $p < 0.001$ (compared with the SHAM-SAL group) #, $p < 0.05$ (significant differences between TBI groups). Data are shown as mean ± standard deviation. Scale bars = 40 μm. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

NAÏVE and SHAM-SAL ($p < 0.001$). In DG, TBI-BIP also showed longer branches (14.44 ± 5.79 μm) than NAÏVE (7.05 ± 1.91 μm; $p = 0.008$). For details, see supplemental Figs. S4–S7 and the triple-point analysis in the supplemental material.

3.11. TBI-BIP group increased brain vascularization

Vessels labeled with tomato lectin (TL) were analyzed 91 days post-trauma.One-way ANOVA revealed significant group differences in vascular expression only in the motor cortex [$F(3,20) = 4.72$, $p < 0.05$] (Fig. 6A–B). TL intensity was higher in TBI-BIP rats (1.03 ± 0.18) compared with SHAM-SAL (0.56 ± 0.19, $p = 0.012$) and TBI-SAL (0.62 ± 0.18, $p = 0.033$; Fig. 6B I).

TL-positive vessel counts also differed in the motor cortex [$F(3,20) = 6.72$, $p < 0.01$], CA1 [$F(3,20) = 4.89$, $p < 0.05$], and CA3 [$F(3,20) = 14.45$, $p < 0.0001$]. The TBI-BIP group showed increased vessel density compared to SHAM-SAL and TBI-SAL in all three regions (motor cortex: 841 ± 135 vs. 545 ± 191 and 536 ± 70; $p < 0.01$; CA1: 786 ± 186 vs. 503 ± 97; $p = 0.01$; CA3: 820 ± 184 vs. 354 ± 124 and 478 ± 190; $p < 0.01$; Fig. 6B II). In CA3, NAÏVE rats (823 ± 98) also exhibited more vessels than SHAM-SAL ($p = 0.0002$) and TBI-SAL ($p = 0.005$).

4. Discussion

Herein, we report that biperiden administration after TBI enhanced vascular remodeling and partially reduced seizure susceptibility and neuroinflammation. Although these effects were modest, biperiden did not worsen any outcome, highlighting its acceptable safety profile. While not strongly neuroprotective, the absence of detrimental effects is important when evaluating its potential therapeutic value.

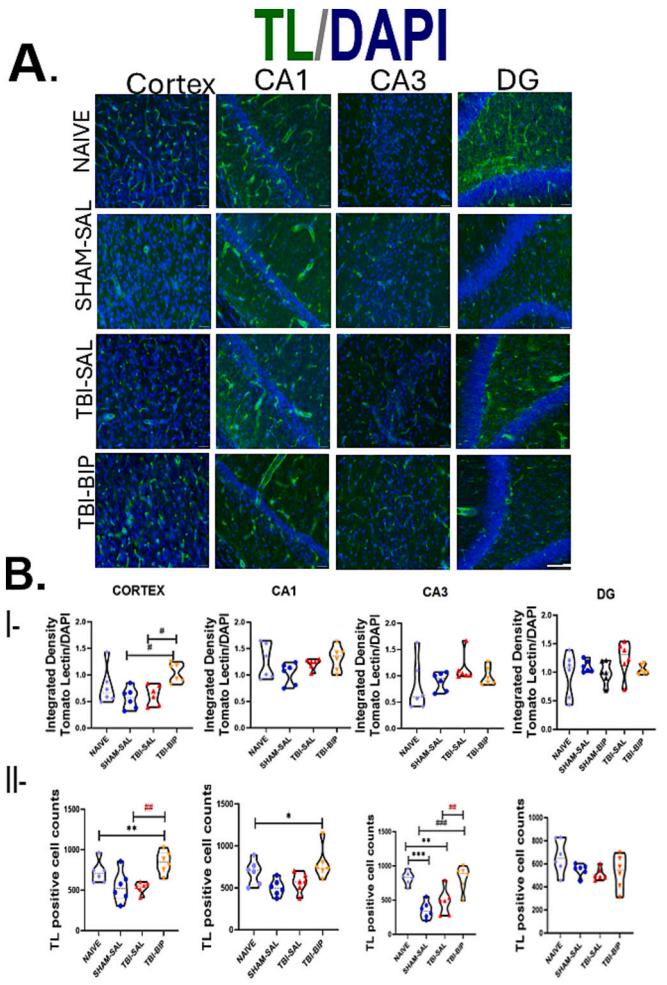


Fig. 6. Tomate Lectina (TL) results. A.Photomicrographs of TL immunofluorescence of all groups' motor cortex and hippocampal regions (CA1, CA3, and DG).TL (green), DAPI (blue). B I– Measure the integrated density of TL, correcting using DAPI, in the cortex, CA1, CA3, and DG. II—TL positive cells count in the cortex, CA1, CA3, and DG. Statistical significance: *, $p < 0.05$; **, $p < 0.01$; ***, $p < 0.001$; ****, $p < 0.001$ (compared with the NAIVE group).#, $p < 0.05$ (compared with the SHAM-SAL group).#, $p < 0.05$; ##, $p < 0.01$; ###, $p < 0.001$; ####, $p < 0.001$ (compared with the SHAM-SAL group). Data are shown as mean ± standard deviation. Scale bars = 40 μm. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Biperiden produced a brief improvement in sensorimotor performance, as observed in the Neuroscore test. In contrast, TBI-SAL rats displayed persistent deficits across the first 10 days, reflecting the known vulnerability of the motor cortex to immediate vascular and neuronal injury following moderate trauma (David et al., 1997). The Barnes Maze probe further demonstrated cognitive impairment in TBI-SAL animals. Their reliance on inefficient trial-and-error navigation—rather than spatial cue strategies used by controls—suggests long-lasting hippocampal dysfunction (Thompson et al., 2005; Bramlett et al., 1997; Fedor et al., 2010). The persistence of memory deficits at 90 days post-injury supports the idea that LFPI causes significant chronic disruption of hippocampal circuits (Qu Yan et al., 2015).Although biperiden can induce amnesia and impair learning in clinical settings—likely through its actions on M1 receptors, which are essential for memory function (Kimura et al., 1999)—our data suggest no lasting cognitive impact. By 30 days post-injury and 20 days after the final biperiden dose, spatial memory performance in the TBI-BIP group was comparable to that of the control group, as reflected in accuracy in locating the target hole (Fig. 2D).

Although LFPI is primarily a TBI model, we observed SRS in one TBI-SAL rat (5 %) beginning at day 61 post-trauma, progressing from type III to IV, suggesting ongoing epileptogenesis (Kalviainen et al., 1998). While based on a single animal, this finding raises important questions about the timeline of PTE development (Kalviainen et al., 1998). Previous studies show that 92–100 % of rats with severe LFPI develop SRS within six months (D'Ambrosio et al., 2005), while other studies mentioned 43–50 % of rats develop epilepsy between 7 weeks and 1 year (Kharatishvili et al., 2006). Thus, our 13-week monitoring window may have been too short to capture the full course of PTE. Because ongoing epileptogenesis contributes to hippocampal deterioration, we used the PTZ challenge to examine seizure threshold and latent hyperexcitability.

The PTZ test revealed clear differences across groups. At 30 mg/kg, TBI rats showed reduced seizure threshold and higher susceptibility, indicating increased PTE risk (Thompson et al., 2005). SHAM animals also displayed seizures, though with longer latency and reduced severity, suggesting that craniotomy alone may temporarily alter cortical excitability (Cole et al., 2011). Importantly, biperiden treatment reduced seizure intensity compared with TBI-SAL animals. This effect may reflect muscarinic receptor antagonism, which can lower excessive cholinergic and glutamatergic activity—key drivers of seizure propagation following TBI (Picciotto et al., 2012). Muscarinic blockade may also dampen inflammatory and vascular responses, thereby stabilizing neural networks and reducing excitotoxicity (Di Lazzaro et al., 2000). Although the treatment did not normalize seizure thresholds, these findings support a modest, dose-limited protective role for biperiden against chemically induced seizures.

Both TBI groups displayed elevated plasma NfL levels 10 days after injury, reinforcing the association between NfL and axonal damage following TBI (Gao et al., 2020; Bacioglu et al., 2016). No differences were observed in plasma T-tau. These results differ from unpublished observations from Braga et al. (2025) (Braga et al., 2025), who reported group differences in NfL and T-tau during the acute phase (24-h), highlighting the importance of temporal resolution in interpreting biomarker dynamics. One possibility is that any biperiden-mediated neuroprotection is transient, consistent with the short-lived functional improvements observed in the Neuroscore on day 3. Another important consideration is the temporal profile of NfL: this biomarker typically peaks within the first 24–72 h after axonal injury and then declines. Thus, NfL levels measured at day 10 likely reflect a later stage in which treatment-related differences may already be attenuated. This temporal shift may obscure early neuroprotective effects. Therefore, future studies should include acute (≤24 h) NfL measurements, to allow direct comparison with Braga et al., and better determine the time course of biperiden's neuroprotective actions.

The preserved NeuN-positive cell count in CA3 and DG in the TBI-BIP group suggests partial neuroprotection by biperiden. In contrast, TBI-SAL rats exhibited pronounced neuronal loss in CA1, CA3, DG, and motor cortex. These findings mirror those of Balança et al. (2016) (Balança et al., 2016), who reported substantial neuronal loss in the temporal cortex and CA3, 7 days after severe LFPI (Balança et al., 2016). Regions showing the earliest and greatest vulnerability likely sustain extensive initial damage that later progresses through secondary degeneration. Chronic inflammation at the primary injury site may further amplify neuronal loss (Balança et al., 2016). Our results suggest that biperiden helps mitigate neuronal loss in CA3, DG, and the motor cortex, but not in the CA1 region (Kumar and Loane, 2012).

TBI induced significant glial remodeling in both microglia and astrocytes, characterized by reduced branching and increased process length. Dysregulated microglial activation can prolong neuroinflammation and exacerbate secondary injury (Rao et al., 2000; Girgis et al., 2016; Viviani et al., 2014). Reduced Iba-1 expression in hippocampal regions may reflect microglial apoptosis or migration to other brain areas following trauma (Lier et al., 2021; White and Vandevord, 2023). Astrocytic reactivity, reflected by increased GFAP expression,

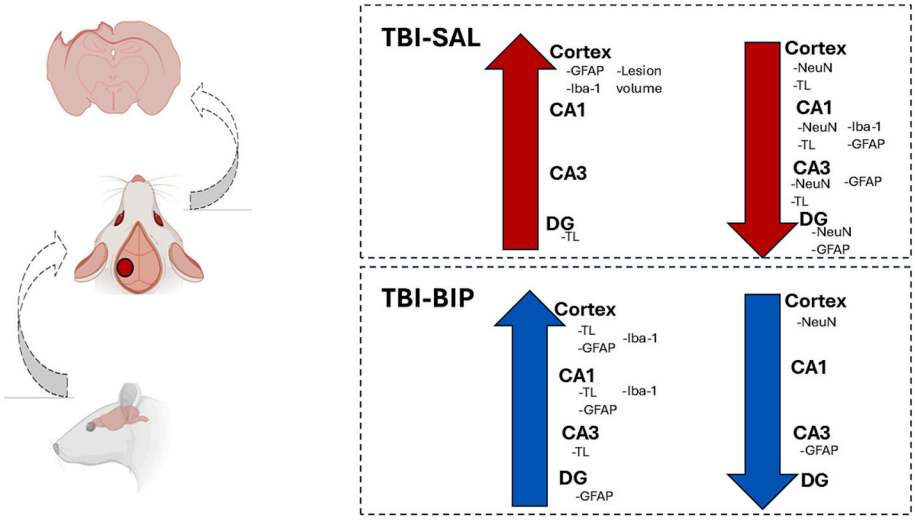


Fig. 7. Summary of Experimental Findings. A comprehensive visual summary of key experimental results, highlighting significant outcomes across histological and immunofluorescence analyses. The figure integrates data on neuronal nuclei (NeuN), microglial and astrocytic responses (Iba-1 and GFAP), and vascular changes (TL).

was still evident at chronic time points, aligning with previous studies (Lee et al., 2018). Unexpectedly, we observed a decrease in GFAP+ astrocytes at the impact site, especially in TBI-SAL animals, suggesting possible astrocyte death, migration, or structural damage to intermediate filaments (Zhou et al., 2020). Biperiden-treated rats displayed a smaller reduction in hippocampal astrocyte density, indicating partial preservation of astrocytic integrity. Although the hippocampus is not directly impacted in LFPI, mechanical forces propagate into deeper regions, causing glial changes (White and Vandevord, 2023). These observations support the idea that biperiden may indirectly modulate glial responses by altering inflammatory or vascular processes. The elevated glial expression seen in SHAM-SAL animals likely reflect craniotomy-induced injury. Disruption of blood vessels during bone-flap removal can trigger delayed secondary cascades, including biochemical processes that promote local cytotoxic edema (Cole et al., 2011). These effects explain why SHAM-SAL animals differed from NAÏVE controls, and support the interpretation of SHAM as a procedural control, not a true baseline.A major finding of this study was the vascular remodeling associated with biperiden treatment. Increased TL expression in the motor cortex of TBI-BIP rats suggests that biperiden may counteract TBI-induced hypoperfusion and promote vascular repair and neovascularization. In contrast, reduced TL expression in TBI-SAL rats may represent progressive deterioration in the perilesional area—a region known to undergo delayed pathological changes over several months (Immonen et al., 2009). Imaging studies by Immonen et al. (2009) showed continuous structural damage for at least six months following TBI, often driven by vasogenic edema (Immonen et al., 2009). By enhancing vascular remodeling, biperiden may help support perfusion and minimize chronic deterioration.

SHAM-SAL rats generally performed better than the TBI groups; however, histological analyses revealed notable vascular and cellular alterations, as well as alterations following the PTZ challenge. These findings emphasize that SHAM animals are not free from injury-related changes. Mechanical trauma from craniotomy may result from the disruption of nerve fibers (Cole et al., 2011). The use of electric drills exacerbates this effect through heat transmission, making them more harmful than manual trephines (Cole et al., 2011), although both methods induced changes consistent with TBI (Cole et al., 2011). This underscores the importance of interpreting SHAM data with caution and recognizing procedural effects when studying TBI.

The observed effects of biperiden are consistent with its well-established safety profile in clinical settings (Benassi et al., 2021). However, a recent clinical study by Foresti et al. (2024) (Foresti et al., 2024) did not detect clear benefits of biperiden in TBI patients, highlighting the need to optimize treatment duration and dosing. Supporting this idea, Benassi et al. (2021) (Benassi et al., 2021) reported a transient antiepileptogenic effect after 17 days of muscarinic blockade with scopolamine in a pilocarpine-induced SE model. This suggests that prolonged biperiden administration may be required to achieve stronger therapeutic benefits.

In summary, biperiden showed beneficial effects, including improved vascular remodeling and partial preservation of neurons in the motor cortex, CA3, and DG (Fig. 7). However, these effects were insufficient to prevent long-term consequences, such as increased seizure susceptibility. The lack of sustained improvement over 91 days suggests that the dose or timing of biperiden administration may need to be optimized to achieve a more robust therapeutic outcome. Additionally, SHAM group data require careful interpretation, as minor surgical interventions may influence both behavioral and cellular responses. Future studies should explore craniotomy-induced glial activation and behavioral changes to better delineate the contribution of procedural effects in experimental TBI, and acknowledge that assessing albumin leakage or blood-brain barrier (BBB) integrity would provide valuable additional insight into biperiden's effects on vascular permeability.

CRediT authorship contribution statement

Viviam Sanabria: Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Conceptualization. **Matheus B. Braga:** Writing – review & editing, Writing – original draft, Methodology. **Simone Romariz:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Conceptualization. **Christiane Gimenes:** Writing – review & editing, Writing – original draft, Methodology, Investigation. **Michele Longoni Calió:** Writing – review & editing, Writing – original draft, Methodology, Investigation. **Maira Licia Foresti:** Writing – review & editing, Writing – original draft, Project administration, Conceptualization. **Luiz Eugênio Mello:** Writing – review & editing, Writing – original draft, Supervision, Resources, Project administration, Conceptualization. **Beatriz M. Longo:** Writing – review & editing, Writing – original draft, Supervision, Conceptualization.

Declaration of competing interest

The authors declare no competing interests.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.expneurol.2025.115626.

Data availability

Data will be made available on request.

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